Zakaria Goumri

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Professional Summary

Backend and AI Developer with 3+ years of experience building scalable backend services, LLM-powered systems, and AI agents. 4th-year Computer Science engineering student at ESI SBA (Computer Systems Engineering). Passionate about DevOps and cloud infrastructure: containers, Kubernetes, CI/CD, and observability.

Professional Experience

Tyrhal Dec 2025 – Present
Backend Developer (DevOps & AI) Remote

- Built and managed the NestJS backend: 50+ endpoints, CQRS, Prisma, PostgreSQL, Redis pub/sub for real-time.
- Built Bull background workers (bookings, notifications, emails, PDF export, translation) and cron schedulers.
- Owned DevOps on DigitalOcean: Docker, GitHub Actions CI/CD to DO Registry for dev, staging, and prod.
- Wrote Kubernetes manifests (API, PostgreSQL, Valkey, MinIO) and ran Trivy/Gitleaks security scans.
- Wrote unit and integration tests with Jest and Testcontainers, plus k6 load tests.
- Designed AI-driven dynamic pricing and an LLM-powered Package Agent, served by FastAPI microservices.

Ooredoo Algeria Sep 2026
IT Intern – Cloud & Infrastructure Ouled Fayet, Algiers

- Worked with the Cloud team on the data center: VMware vSphere, SAN zoning, VDI, and network setup.
- Shadowed the Database team: Oracle, PostgreSQL, MongoDB administration, backups, access control, monitoring.
- Shadowed the DevOps team: Docker, Kubernetes orchestration, Azure DevOps CI/CD, Grafana monitoring.
- Explored the on-premise AI stack: self-hosted Qwen on 6 GPUs, LiteLLM gateway, Langfuse observability.

Portals Feb 2025 – May 2025
Backend Developer Remote

- Designed scalable backend architecture using NestJS and PostgreSQL, with real-time notifications via SSE.
- Developed analytics APIs for dashboards and reporting systems.

Technical Skills

Languages: TypeScript, JavaScript, Go, Python
Backend: NestJS, FastAPI, Microservices, CQRS, Background Queues, Redis Pub/Sub, Kafka, GraphQL, gRPC, WebSockets, SSE
AI / ML: LLMs, AI Agents, RAG, LangChain, Embeddings, Vector Databases
Databases: PostgreSQL, MongoDB, Redis, Elasticsearch, Neo4j, Prisma, TypeORM
Security: OAuth 2.0, PKCE, JWT, RBAC, AWS SigV4, IAM Policies
DevOps & Cloud: Docker, Kubernetes (K8s), DigitalOcean, GitHub Actions, Linux, Nginx, Grafana
Spoken Languages: Arabic (Native), English (Professional), French (Professional)

Projects

Vexo – S3-Compatible Object Store *Go, bbolt, Vue 3, Docker*

- Built a single-node object store exposing a real S3 HTTP API, compatible with SigV4 clients (aws-cli, boto3).
- Implemented AWS SigV4 authentication and an IAM-style JSON policy engine with users, groups, and access keys.
- Built lifecycle rules that move objects across hot, infrequent, and cold (gzip) tiers using a background scanner.
- Shipped a Vue 3 web console, a custom binary TCP protocol, and a Dockerized setup with Docker Compose.

NemoBase – Database Engine From Scratch *Go*

- Building a relational database from scratch, starting with a copy-on-write B+tree storage engine.
- Implemented page-based node layout and crash-safe atomic writes; next: KV store, indexes, transactions, query language.

Stamp – OAuth 2.0 Authorization Server for NestJS *TypeScript, NestJS*

- Built a reusable NestJS module that turns any app into an OAuth 2.0 authorization server.
- Implemented the Authorization Code flow with PKCE, and a pluggable client store for any database.

Education

Higher National School of Computer Science (ESI SBA) 2023 – 2028
Engineering Degree in Computer Science, Computer Systems Engineering (ISI) Sidi Bel Abbès, Algeria

Leadership & Community

- Google Developer Student Club (GDSC SBA) – Volunteer / Organizer
- Hackathons: 1st Place at Qurbani (AI system for donation transparency), 2nd Place at Caravane Assurini